

17 Related Substances

Ammonia solution.

18 Comments

Ammonium chloride has the ability to cross the red blood cell membrane, and a solution that is isotonic to blood will still cause hemolytic rupture because it acts as a hypotonic solution.

A specification for ammonium chloride is contained in the Food Chemicals Codex (FCC).⁽¹¹⁾

The EINECS number for ammonium chloride is 235-186-4. The PubChem Compound ID (CID) for ammonium chloride is 25517.

19 Specific References

- 1 Gottardi W *et al.* N-Chlorotaurine and ammonium chloride: an antiseptic preparation with strong bactericidal activity. *Int J Pharm* 2007; 335: 32–40.
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- 6 Zhu RS *et al.* Sublimation of ammonium salts: a mechanism revealed by a first-principles study of the NH₄Cl system. *J Phys Chem* 2007; 111: 13831–13838.

- 7 NIOSH Pocket Guide to Chemical Hazards (DHHS/NIOSH 97-140) 1997; 16.
- 8 Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 231.
- 9 Boyd EM, Seymour KGW. Ethylene diamine dihydrochloride. II. Untoward toxic reactions. *Exp Med Surg* 1946; 4: 223–227.
- 10 Smeets P. Ammonium chloride [and water treatment]. *Tribune de l'Eau* 1994; 47(570): 26–29.
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20 General References

- Ingham JW. The apparent hydration of ions. III. The densities and viscosities of saturated solutions of ammonium chloride in hydrochloric acid. *J Chem Soc* 1929; 2059–2067.
- Kumaresan R *et al.* Simultaneous heat and mass transfer studies in drying of ammonium chloride in fluidized bed dryer. *Process Plant Eng* 2007; 25(3): 60–66.

21 Author

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22 Date of Revision

27 February 2009.

Ascorbic Acid

1 Nonproprietary Names

BP: Ascorbic Acid
JP: Ascorbic Acid
PhEur: Ascorbic Acid
USP: Ascorbic Acid

2 Synonyms

Acidum ascorbicum; C-97; cevitamic acid; 2,3-didehydro-L-threo-hexono-1,4-lactone; E300; 3-oxo-L-gulofuranolactone, enol form; vitamin C.

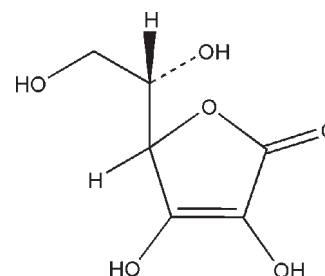
3 Chemical Name and CAS Registry Number

L-(+)-Ascorbic acid [50-81-7]

4 Empirical Formula and Molecular Weight

C₆H₈O₆ 176.13

5 Structural Formula



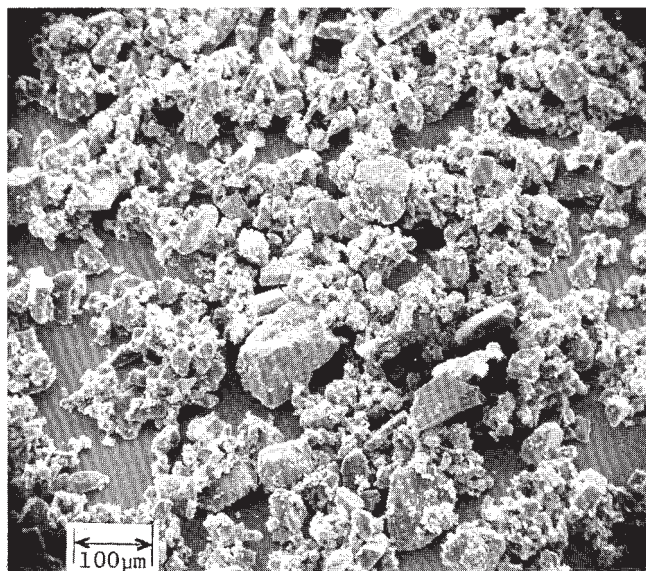
6 Functional Category

Antioxidant; therapeutic agent.

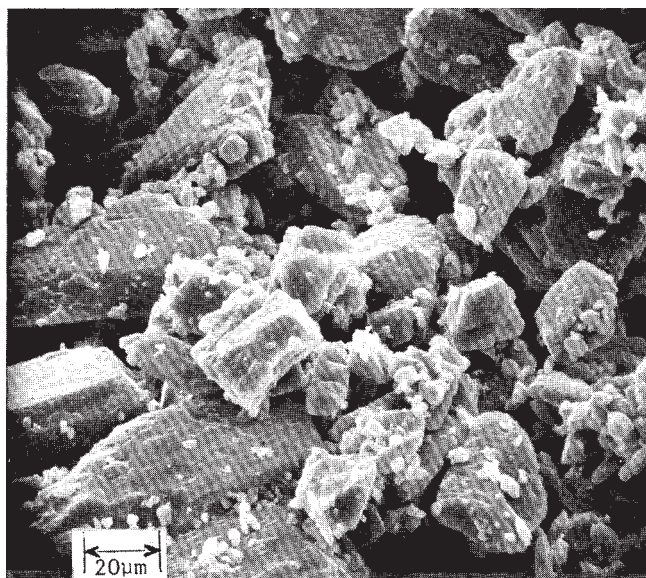
7 Applications in Pharmaceutical Formulation or Technology

Ascorbic acid is used as an antioxidant in aqueous pharmaceutical formulations at a concentration of 0.01–0.1% w/v. Ascorbic acid has been used to adjust the pH of solutions for injection, and as an adjunct for oral liquids. It is also widely used in foods as an antioxidant. Ascorbic acid has also proven useful as a stabilizing agent in mixed micelles containing tetrazepam.⁽¹⁾

SEM 1: Excipient: ascorbic acid usp (fine powder); manufacturer: Pfizer Ltd; lot no.: 9A-3/G92040-CO 146; magnification: 120×; voltage: 20 kV.



SEM 2: Excipient: ascorbic acid usp (fine powder); manufacturer: Pfizer Ltd; lot no.: 9A-3/G92040-CO 146; magnification: 600×; voltage: 20 kV.



8 Description

Ascorbic acid occurs as a white to light-yellow-colored, nonhygroscopic, odorless, crystalline powder or colorless crystals with a sharp, acidic taste. It gradually darkens in color upon exposure to light.

9 Pharmacopeial Specifications

See Table I.

10 Typical Properties

Acidity/alkalinity pH = 2.1–2.6 (5% w/v aqueous solution)

Density (bulk)

0.7–0.9 g/cm³ for crystalline material;

0.5–0.7 g/cm³ for powder.

Density (particle) 1.65 g/cm³

SEM 3: Excipient: ascorbic acid usp (fine granular); manufacturer: Pfizer Ltd; lot no.: 9A-2/G01280-CO 148; magnification: 120×; voltage: 20 kV.

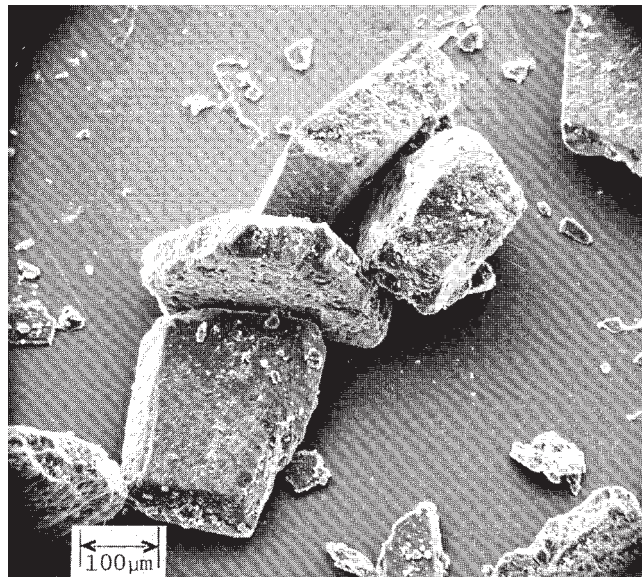


Table I: Pharmacopeial specifications for ascorbic acid.

Test	JP XV	PhEur 6.3	USP 32
Identification	+	+	+
Characters	—	+	—
Specific rotation (10% w/v solution)	+ 20.5° to + 21.5°	+ 20.5° to + 21.5°	+ 20.5° to + 21.5°
Residue on ignition	≤0.1%	—	≤0.1%
pH	2.2–2.5	2.1–2.6	—
Sulfated ash	—	≤0.1%	—
Copper	—	≤5 ppm	—
Heavy metals	≤20 ppm	≤10 ppm	≤0.002%
Loss on drying	≤0.20%	—	—
Iron	—	≤2 ppm	—
Oxalic acid	—	+	—
Related substances	—	+	—
Appearance of solution	+	+	—
Assay	≥99.0%	99.0–100.5%	99.0–100.5%

Table II: Solubility of ascorbic acid.

Solvent	Solubility at 20°C
Chloroform	Practically insoluble
Ethanol	1 in 50
Ethanol (95%)	1 in 25
Ether	Practically insoluble
Fixed oils	Practically insoluble
Glycerin	1 in 1000
Propylene glycol	1 in 20
Water	1 in 3.5

Density (tapped)

1.0–1.2 g/cm³ for crystalline material;

0.9–1.1 g/cm³ for powder.

Density (true) 1.688 g/cm³

Dissociation constant

pK_{a1} = 4.17;

pK_{a2} = 11.57.

Melting point 190°C (with decomposition)

Moisture content 0.1% w/w

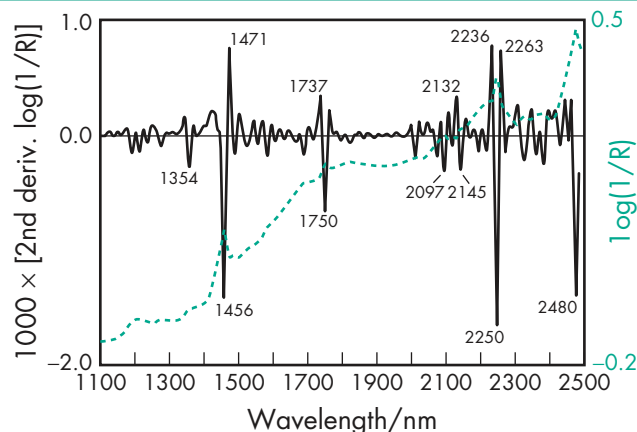


Figure 1: Near-infrared spectrum of ascorbic acid measured by reflectance.

NIR spectra see Figure 1.

Solubility see Table II.

11 Stability and Storage Conditions

In powder form, ascorbic acid is relatively stable in air. In the absence of oxygen and other oxidizing agents it is also heat stable. Ascorbic acid is unstable in solution, especially alkaline solution, readily undergoing oxidation on exposure to the air.^(2,3) The oxidation process is accelerated by light and heat and is catalyzed by traces of copper and iron. Ascorbic acid solutions exhibit maximum stability at about pH 5.4. Solutions may be sterilized by filtration.

The bulk material should be stored in a well-closed nonmetallic container, protected from light, in a cool, dry place.

12 Incompatibilities

Incompatible with alkalis, heavy metal ions, especially copper and iron, oxidizing materials, methenamine, phenylephrine hydrochloride, pyrilamine maleate, salicylamide, sodium nitrite, sodium salicylate, theobromine salicylate, and picotamide.^(4,5) Additionally, ascorbic acid has been found to interfere with certain colorimetric assays by reducing the intensity of the color produced.⁽⁶⁾

13 Method of Manufacture

Ascorbic acid is prepared synthetically or extracted from various vegetable sources in which it occurs naturally, such as rose hips, blackcurrants, the juice of citrus fruits, and the ripe fruit of *Capsicum annuum* L. A common synthetic procedure involves the hydrogenation of D-glucose to D-sorbitol, followed by oxidation using *Acetobacter suboxydans* to form L-sorbose. A carboxyl group is then added at C1 by air oxidation of the diacetone derivative of L-sorbose and the resulting diacetone-2-keto-L-gulonic acid is converted to L-ascorbic acid by heating with hydrochloric acid.

14 Safety

Ascorbic acid is an essential part of the human diet, with 40 mg being the recommended daily dose in the UK⁽⁷⁾ and 60 mg in the USA.⁽⁸⁾ However, these figures are controversial, with some advocating doses of 150 or 250 mg daily. Megadoses of 10 g daily have also been suggested to prevent illness although such large doses are now generally considered to be potentially harmful.^(9–11)

The body can absorb about 500 mg of ascorbic acid daily with any excess immediately excreted by the kidneys. Large doses may cause diarrhea or other gastrointestinal disturbances. Damage to the teeth has also been reported.⁽¹²⁾ However, no adverse effects have been reported at the levels employed as an antioxidant in foods, beverages,⁽¹³⁾ and pharmaceuticals. The WHO has set an

acceptable daily intake of ascorbic acid, potassium ascorbate, and sodium ascorbate, as antioxidants in food, at up to 15 mg/kg body-weight in addition to that naturally present in food.⁽¹⁴⁾

LD₅₀ (mouse, IV): 0.52 g/kg⁽¹⁵⁾

LD₅₀ (mouse, oral): 3.37 g/kg

LD₅₀ (rat, oral): 11.9 g/kg

15 Handling Precautions

Ascorbic acid may be harmful if ingested in large quantities and may be irritating to the eyes. Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection and rubber or plastic gloves are recommended.

16 Regulatory Status

GRAS listed. Accepted for use as a food additive in Europe. Included in the FDA Inactive Ingredients Database (inhalations, injections, oral capsules, suspensions, tablets, topical preparations, and suppositories). Included in medicines licensed in the UK. Included in the Canadian List of Acceptable Non-medicinal Ingredients.

17 Related Substances

Ascorbyl palmitate; erythorbic acid; sodium ascorbate.

18 Comments

Many dosage forms for ascorbic acid have been developed for its administration to patients, including microencapsulation.⁽¹⁶⁾

A specification for ascorbic acid is contained in the Food Chemicals Codex (FCC).⁽¹⁷⁾

The EINECS number for ascorbic acid is 200-066-2. The PubChem Compound ID (CID) for ascorbic acid is 5785.

19 Specific References

- 1 Hammad MA, Muller BW. Solubility and stability of tetrazepam in mixed micelles. *Eur J Pharm Sci* 1998; 7: 49–55.
- 2 Hajratwala BR. Stability of ascorbic acid. *STP Pharma* 1985; 1: 281–286.
- 3 Tountou E *et al.* Ascorbic acid in aqueous solution: bathochromic shift in dilution and degradation. *Int J Pharm* 1992; 78: 85–87.
- 4 Botha SA *et al.* DSC screening for drug–drug interactions in polypharmaceuticals intended for the alleviation of the symptoms of colds and flu. *Drug Dev Ind Pharm* 1987; 13: 345–354.
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- 8 Subcommittee on the tenth edition of the RDAs, Food and Nutrition Board, Commission on Life Sciences. National Research Council. *Recommended Dietary Allowances*, 10th edn. Washington, DC: National Academy Press, 1989.
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the joint FAO/WHO expert committee on food additives. *World Health Organ Tech Rep Ser* 1974; No. 539.

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20 General References

- Abramovici B *et al.* [Comparative study of the tableability of different grades of vitamin C.] *STP Pharma* 1987; **3**: 16–22[in French].
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- Krishna G *et al.* Development of a parenteral formulation of an investigational anticancer drug, 3-aminopyridine-2-carboxaldehyde thiosemicarbazone. *Pharm Dev Technol* 1999; **4**: 71–80.

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Seta Y *et al.* Preparation and pharmacological evaluation of Captopril sustained-release dosage forms using oily semisolid matrix. *Int J Pharm* 1988; **41**: 255–262.

21 Author

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22 Date of Revision

10 February 2009.

Ascorbyl Palmitate

1 Nonproprietary Names

BP: Ascorbyl Palmitate

PhEur: Ascorbyl Palmitate

USP-NF: Ascorbyl Palmitate

2 Synonyms

L-Ascorbic acid 6-palmitate; ascorbylis palmitas; E304; 3-oxo-L-gulofuranolactone 6-palmitate; vitamin C palmitate.

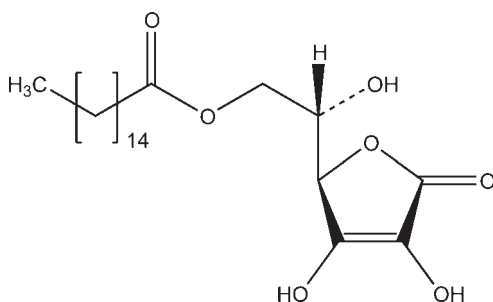
3 Chemical Name and CAS Registry Number

L-Ascorbic acid 6-hexadecanoate [137-66-6]

4 Empirical Formula and Molecular Weight

C₂₂H₃₈O₇ 414.54

5 Structural Formula



6 Functional Category

Antioxidant.

7 Applications in Pharmaceutical Formulation or Technology

Ascorbyl palmitate is primarily used either alone or in combination with alpha tocopherol as a stabilizer for oils in oral pharmaceutical formulations and food products; generally 0.05% w/v is used. It may also be used in oral and topical preparations as an antioxidant for drugs unstable to oxygen. The combination of ascorbyl palmitate with alpha tocopherol shows marked synergism, which increases the effect of the components and allows the amount used to be reduced.

The solubility of ascorbyl palmitate in alcohol permits it to be used in nonaqueous and aqueous systems and emulsions.

8 Description

Ascorbyl palmitate is a practically odorless, white to yellowish powder.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for ascorbyl palmitate.

Test	PhEur 6.0	USP32–NF27
Identification	+	+
Appearance of solution	+	—
Melting range	—	107–117°C
Specific rotation (10% w/v in methanol)	+21° to +24°	+21° to +24°
Loss on drying	≤ 1.0%	≤ 2.0%
Residue on ignition	—	≤ 0.1%
Sulfated ash	≤ 0.1%	—
Heavy metals	≤ 10 ppm	≤ 0.001%
Assay (dried basis)	98.0–100.5%	95.0–100.5%